Lab 2 - Exercise

1. Continuously monitor the state of the switches and update the corresponding LEDs. Whenever a switch is turned ON, the corresponding LED should turn ON. Whenever a switch is turned OFF, the corresponding LED should turn OFF. Any change in the state of a switch should be immediately reflected on its corresponding LED.
2. Control an LED using a switch. Each time the switch is activated, the state of the LED should toggle. The LED should change from ON to OFF or OFF to ON with every switch activation.
3. Generate a running LED pattern. Initially, only P1.0 should be illuminated. The illuminated LED should move one position at a time towards P1.7. After reaching P1.7, the pattern should restart from P1.0 and continue indefinitely.
4. Determine whether an 8-bit unsigned number stored in the accumulator is odd or even. If the number is even, turn ON the 1st LED. If the number is odd, turn ON the last LED. Ensure that only one LED is ON at any time to indicate the result.
5. Perform addition, subtraction, multiplication, and division of two 8-bit numbers. Store the results in different register banks and indicate arithmetic overflow or carry conditions using LEDs.
6. Compare two 8-bit numbers. If the first number is greater, turn ON one LED; if the second number is greater, turn ON another LED; if both numbers are equal, turn ON a third LED.
7. Perform AND, OR, and XOR operations on two 8-bit numbers. Store the two input numbers in memory locations and perform each logical operation separately. Store the resulting values in designated memory locations for verification.
8. Exchange the contents of two registers. After the exchange operation, increment the value of one register and decrement the value of the other register. Store the final values for verification.
9. Monitor the accumulator value and indicate its status using LEDs. If the accumulator contains zero, turn ON one LED; if it contains a non-zero value, turn ON another LED. Also monitor bit 7, and whenever the bit is set, clear the bit and turn ON a third LED.